

Does gender matter for aid project performance? The case of Asian Development Bank.

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The performance of aid projects matters for development. Yet what shapes performance, is insufficiently understood. According to interdisciplinary research on organizational management and bureaucratic representation, the gender of operational leaders in complex organizations can have a discernible effect on all aspects of the project cycle. To this end, I seek to test whether the gender of aid project staff matters in explaining variation in project performance. I use a self-collected novel dataset from the Asian Development Bank to discern whether project staff's gender (women vs men project leaders) is associated with project performance. I use multinomial-ordered logistic regressions to test the association of the gender of project managers with project performance measures of relevance, effectiveness, efficiency, and sustainability. I find that projects led by women are, on average, associated with higher performance ratings on dimensions of efficiency. Significant and positive association with the relevance criterion is subject to model specification. However, I do not see an association with the effectiveness and sustainability criterion of project performance. This paper contributes to the literature by estimating the differential impact of women vs men project managers on disaggregated performance measures. It also lays down potential mechanisms and future research discourse.

Keywords: gender; aid-organization; development; Asian Development Bank; project performance

Introduction

The impact of women's leadership on firm performance and organizational practices has been widely studied in the finance, economics, management, and development literature. Recently, the representational bureaucracy theory has been applied to international organizations to study the impact of women's leadership on gender mainstreaming in the organization. I use tools from the above-mentioned interdisciplinary literature to examine the role of staff identity in international

organizations. Specifically, this article seeks to better understand the effects of gender of staff in operational matters – specifically how gender identity shapes outcomes of interest at the project level. I use the case of a regional development bank – the Asian Development Bank (ADB) to ask a central question – When women manage aid projects, are there empirically observable differences between the performance outcomes between men and women?

Empirically, I utilize novel individual and project-level data from 1998-2020 on all completed projects at ADB, and the gender of thousands of ADB staff members in positions of authority - Country Directors, Vice Presidents, Director Generals, project leaders¹, and team members. Evidence from statistical results reveals that women officers are less represented in managerial roles at ADB. The main finding of the paper is that women leaders are significantly associated with an improvement in the relevance and efficiency criterion of project performance, but not the effectiveness, and sustainability criteria.

This investigation contributes to the literature in the following ways: First, it expands on the development literature to include gender as an associated factor for project performance. It expands on the organizational and financial economics literature to move away from corporate entities and look at an aid organization. It also expands on the women's representation theory to include one systemically important regional development bank: the Asian Development Bank. Second, it dissects different aspects of project performance to understand which aspects are impacted by the presence of women on those projects. Research on aid organizations like the World Bank (WB) and

¹ ADB uses the terms project officers and team leaders interchangeably. This paper does the same. Sometimes, project managers are also used in lieu of project officers and/or leaders.

ADB has only utilized one measure of project performance – the overall project outcome (Denizer et al., 2013; Feeny & Vuong, 2017). This study considers the relevance, efficiency, effectiveness, and sustainability measures of project performance separately. My contribution expands the literature to levels of analysis and organizations that have not been studied before in this context. The findings from this study present mixed results on how women are associated with aid-project performance. I highlight theories as to why that might be the case.

This paper proceeds in five steps. The first is to review the interdisciplinary research on women's leadership and identify gaps in the literature. The second is to explain the selection of ADB, its goals, and undertakings as the largest regional development bank. Third, I discuss the research design utilizing individual-level data on thousands of ADB staff members working on projects between 1998-2020, project-level characteristics, and outcomes. Fourth, I present evidence from a range of statistical tests that aim to identify the impact of women on project performance. Fifth, I present robustness checks. Finally, I conclude by highlighting the main findings, policy implications, and future research discourse.

Section 1 – Literature review

What ensures a project's success? The development literature has tried to delve into various factors that could make a project successful. For example, Kilby (2015) uses the World Bank project data to show that longer preparation time increases the odds of success of a World Bank project. He also finds that preparation has more impact in

economically vulnerable countries. Similarly, Khwaja (2009) uses the infrastructure project data of Northern Pakistan to show that project design is key for successful community-based projects. Specifically, he finds that changes in project complexity, community participation, and return distribution can lead to improved project outcomes. Shin et al. (2017) use the World Bank project evaluation data to study the type of implementing partnerships and their impact on project outcomes. They find that projects implemented by non-state actors are likely to be more successful relative to state actors. When making decisions, planning, and implementing projects, and information sharing, non-state actors act with more transparency relative to state actors. That in turn, allows for progress to be monitored more easily. Denizer et al. (2013) investigate macro and micro correlates of aid-financed development project outcomes, using data from over 6000 World Bank projects evaluated between 1983 and 2011. They find that country-level factors such as the quality of institutions are strongly correlated with project outcomes. They also show that micro-project level characteristics such as project preparation time, project size length, funding, etc. are strongly correlated with the success of the project. However, the closest the literature comes to talking about how characteristics of staff matter for project outcomes has been to establish a strong correlation between World Bank project manager quality and project outcomes (Denizer et al., 2013). Very recently Heinzel & Liese (2021) showed that country-specific and sector-specific knowledge of World Bank project managers (known as TTLs at the World Bank) matter for project performance. To my knowledge, however, how women matter for project performance in aid -organizations has not been studied.

There is interdisciplinary literature that we can draw from that shows mixed evidence on how women matter (relative to men) for firm outcomes, political preferences and outcomes, and gender-based outcomes. In public policy, for example,

Cassan & Vandewalle (2021) find that when women are elected for local government positions in India, policies change towards the preferences of women and also towards preferences of the lower caste community, thus altering the distribution of policy, not only along the gender identity lines but also towards the caste dimension. Duflo (2005) finds that the election of women via reservation brings power to a group of relatively inexperienced and less-educated groups. However, there is no evidence that this comes at the expense of the quality of decision-making. In the recent science literature, Koning et al. (2021) show that women, even though they engage in less patenting than men, they are 35% more likely than males to focus on women's health. Flabbi et al. (2019) find that the impact of female leadership on firm performance increases with the share of female workers. Additionally, the performance of a firm led by a female CEO increases with the share of female workers. They theorize that female CEOs are better at processing information about female workers' productivity. Female CEOs improve the allocation of female workers across tasks, improving firm performance overall. Adams & Ferreira (2009) find that male and female directors in corporate firms behave differently with respect to attendance - with women less likely to have attendance problems as compared to men. These findings are in line with the representative bureaucracy theory that gender representation show strong correlations between the gender identity of the bureaucrat and its positive impact on represented clientele (Davies et al., 2020; Dhillon & Meier, 2020; Keiser et al., 2002; Riegle-Crumb & Humphries, 2012; Wilkins, 2006). Global governance scholars have more recently used bureaucratic representation theory to understand the effects of passive representation of women in international organizations (Badache, 2020; Christensen, 2020; Eckhard, 2020; Ege, 2017, 2020, p. 20; Haack et al., 2020; Hooghe, 1999; Parízek, 2017; Reinalda, 2013; Thorvaldsdottir et al., 2021) and specifically in aid organizations

(Heinzel, 2022; Heinzel & Liese, 2021; Weaver et al., 2022). In some other cases, there is no effect of the gender of director/managers on firm performance, the gender of mayors on political outcomes (see Campa, 2011; Ferreira & Gyourko, 2014; Rose, 2007). Campa (2011) points towards the mechanism (for no effect) that there are constraints to women being promoted to executive positions and hence, having influence over policymaking.

Research on how gender matters for policy outcomes such as female educational attainment, improving the lives of the lower caste population, changing perceptions towards female leaders, and improving firm performance is documented (see for example - Adams & Ferreira, 2009; Cassan & Vandewalle, 2021; Duflo, 2005; Flabbi et al., 2019). Preferences, choices, and the way women work differ from men in these settings have also been documented (see for example - Adams & Ferreira, 2009; Cassan & Vandewalle, 2021; Koning et al., 2021). But what do we know about how gender matters for outcomes in international aid organizations? International aid organizations are different from private firms in their goals (Ravallion, 2016). Private firms are designed and aim to achieve profits, whereas international aid organizations carry out developmental aid projects that directly aim to improve the economic, social, and political conditions of target communities. Therefore, one could safely assume that an average staff working at private firms would be inherently different from staff working at international aid organizations. The type of people who work at international aid organizations are more driven by intrinsic motivation and hence their actions might be different from staff at private firms. Research by Kamas et al. (2008) finds that “Women tend to give more than men, and a two-person male-female team gives more than the expected sum of what they would give playing alone”. Thus, identity characteristics such as gender can play an important role in organization dynamics, and

more so in international aid organizations where intrinsic motivations, backgrounds, and passion for certain tasks can exacerbate roles and preferences leading to differences in project outcomes. The evidence on how gender matters in international organizations for gender mainstreaming has been very recent – and is limited to research on the World Bank (Heinzel, 2022). However, to my knowledge, how gender matters for performance in regional development banks has not been studied.

Therefore, this paper seeks to contribute to the literature in the following ways. First, it quantitatively measures the association of gender with project outcomes in a regional development bank. Most of what we know about projects and their staff, project performance, and factors that lead to differences in performance comes from the International Monetary Fund (IMF), the European Union (EU), the United Nations (UN), and more specifically the World Bank (see for example, Blackmon, 2021; Heinzel et al., 2021; Pollack & Hafner-Burton, 2010; Weaver, 2010; Weaver et al., 2022). However, research on ADB is almost non-existent – and where it exists, is mostly descriptive (Feeny & Vuong, 2017). Moreover, while WB and ADB might be similar in their operational structure, they are inherently different in their staff demographics. The culture, social, gender norms, and hierarchies, in the regions from where the staff in both these organizations are hired vary significantly. These fundamental differences might cause a difference in outlook, behaviours, and perceptions of staff and may vary significantly by gender (cite?). Therefore, results from the WB might not be generalizable to regional development banks such as ADB. Moreover, this paper moves beyond the aggregate measure of project performance – the overall project performance measure (Denizer et al., 2013; Feeny & Vuong, 2017).

Section 2 – Background on ADB and operations

ADB is a multilateral regional development bank and provides loans, technical assistance, grants, and equity investments to its 69 member countries. It aims to promote social and economic development in Asia and the Pacific. All ADB projects that incur expenditures undertaken by ADB and are regarded as completed are “self-evaluated by entities and persons responsible for the design and implementation of the intervention”(Vijayaraghavan, 2020, pg 7). Projects are evaluated by project officers and other project members once they are deemed completed. Inputs from borrowers, executing agencies, and beneficiaries go into the estimation of project performance. Two types of reports are furnished with detailed evaluation measures - namely - Project Completion Reports (PCRs) or Technical Assistance Completion Reports (TCRs).

A project lifecycle begins with approval of the project by the Vice-President and the Director. The Project officer is then assigned to the project by the country director based on their sector and educational expertise. Project officers determine targets and outputs, design activities to meet these targets, decide on the gender components, hire consultants, confer with various stakeholders, implement, and monitor the progress of the project. Project officers also play a vital role in evaluating project performance.

The performance of projects at ADB is evaluated by measuring the project’s relevance, effectiveness, efficiency, and sustainability. Relevance is measured by the project’s consistency of proposed outcomes with the country's development priorities and the appropriateness of the design, instruments, and planned implementation of the project based on the country’s needs. Relevance ratings are measured as highly relevant, relevant, less than relevant, and irrelevant. The effectiveness criterion is evaluated based on whether and how the targets and outcomes, as defined in the design and monitoring

framework at the beginning of the project, are achieved. Effectiveness considers staff capabilities, implementation process, government's macro and sector policies, maintenance procedures, and availability of inputs. Effectiveness ratings are measured as highly effective, effective, less than effective, and ineffective. Efficiency is the measure of how well resources were used during the implementation of the project towards achieving outcomes. "It is measured in economic terms because it examines whether the project was an efficient use of resources for the country and/or society (not merely for the operating entity)." (Asian Development Bank, 2016). Efficiency is measured as highly efficient, efficient, less than efficient, and inefficient. The sustainability criteria looks at whether and to what extent the outcomes of the project can be sustained over the project's economic lifetime. It is measured as highly sustainable, sustainable, less than sustainable, unsustainable. The criteria are given a point score of 0 to 3 where 0 is the worst performing and 3 is the best performing. These four criteria are given a 25% weight each to aggregate to an overall assessment rating. The overall assessment is rated as unsuccessful, less than successful, successful, and highly successful (Asian Development Bank, 2016).

Section 3 – Research Design

Data

Data for this paper was collected for approximately 4500 projects that were approved or closed between 1998 and 2020. Publicly available project documents -namely concept notes, reports or recommendations to the President, technical assistance reports, project administration manuals, grant agreements, initial poverty and social analysis, technical assistance completion reports, and project completion reports were read in detail. Information on the project's staff at the start and end of the project, project design

activities, funding, timelines, and project outcomes were manually entered into a Qualtrics survey form. The final dataset has 592 variables on 3471 projects - their staff and outcomes.

Dependent Variables

The dependent variables are the performance ratings on relevance, effectiveness, efficiency, and sustainability. Ratings are on an ordered scale of 0-3, where 0 represents irrelevant, ineffective, inefficient, unlikely sustainable. 1 represents less than relevant, less than effective, less than efficient, and less likely sustainable. 2 represents relevant, effective, efficient, and likely sustainable. 3 is coded for highly relevant, highly effective, highly efficient, and more than likely sustainable. These ratings have been manually collected from PCRs and TCRs. Figure 1 shows the distribution of the four ratings.

Independent Variables

The main independent variable is the gender of the project officer. To determine gender and nationality, I use an application called genderize.io which is 97% accurate in predicting the man and woman gender based on how people identify themselves on social media accounts. I hand-searched for gender when genderize.io came up with an unknown gender. While the first names of all project officers were available, those of staff members were not. Therefore, I google searched the first names of all staff members using the query “Last name Asian Development Bank”. The query either came up with the full name that I used genderized.io on, or a picture, or the terms “Mr.” and

“Mrs.”, or a combination of all. I used these to determine the gender of the staff members. The search was conducted for 2249 unique staff at ADB including executive staff, project officers, and project team members. Figure 2 shows the distribution of project officers based on their gender identity. Approximately 30% of project officers are women and 70% are men. Figure 3 shows the distribution of ratings by the gender of the project officer.

Control variables

I control for project-level characteristics like funding, project type, and project agenda. There are three project agendas – environmentally sustainable growth, inclusive economic growth, and regional integration. Each project is classified as having one or more of these agendas. Project type is classified as Grants, Technical Assistance, Loans, or some combination of these three. I also control for individual-level characteristics such as the position of the officer. There are 342 positions in ADB. I coded them to get 67 unique positions. Project officer nationality is not included in the analysis as it is highly correlated with the country of operation. Country-level controls include the log of gross domestic product (GDP) and the log of population. These controls have been shown in the literature to be significant predictors of project performance.

Summary Statistics

Table 1 shows summary statistics of variables included in the analysis. Reliability, effectiveness, efficiency, and sustainability are outcome variables. There is around 50% missing data on the outcome variables. The independent variable – the gender of the project officer has no missing data. Officer quality is calculated using the method by

Denizer et al. 2013. Quality is determined by average performance rating on projects other than the project in question. Since this variable is calculated using performance rating, it is highly correlated to the outcome measures of efficiency and effectiveness. Therefore, it is not included in the analysis. Table 2 shows project performance ratings by gender of the project officer.

Model Specification

To test whether women officers produce different outcomes of relevance, effectiveness, efficiency, and sustainability relative to men officers, I regress project relevance, effectiveness, efficiency, and sustainability (respectively) on project officer gender. I use multinomial logit regressions with robust standard errors to account for the ordered categorical nature of the performance measures. I include country (c), sector (s), approval year (t(a)), and closing year (t(c)) year fixed effects. I also use entropy matching to match continuous and ordered control variables for more robust estimations.

$$\text{Project Relevance}_{jc} = \alpha + \beta_1 \text{GenderPO}_{jcs} + \beta_2 X_{jcs} + \mu_{ct(a)} + \pi_s + \sigma_{t(c)} + \varepsilon_{jcst} \quad (\text{i})$$

$$\text{Project Effectiveness}_{jc} = \alpha + \beta_1 \text{GenderPO}_{jcs} + \beta_2 X_{jcs} + \mu_{ct(a)} + \pi_s + \sigma_{t(c)} + \varepsilon_{jcst} \quad (\text{ii})$$

$$\text{Project Efficiency}_{jc} = \alpha + \beta_1 \text{GenderPO}_{jcs} + \beta_2 X_{jcs} + \mu_{ct(a)} + \pi_s + \sigma_{t(c)} + \varepsilon_{jcst} \quad (\text{iii})$$

$$\text{Project Sustainability}_{jc} = \alpha + \beta_1 \text{GenderPO}_{jcs} + \beta_2 X_{jcs} + \mu_{ct(a)} + \pi_s + \sigma_{t(c)} + \varepsilon_{jcst} \quad (\text{iv})$$

Section 4 – Results

Results for the model specification above are presented in Table 3, Table 4, Table 5, and Table 6. For each table, model 1 measures the association of women officers and project outcomes (of relevance, effectiveness, efficiency, and sustainability) without controls or fixed effects. Model 2 includes country, sector, and year (approval year and

closing year) fixed effects. Model 3 adds project-level characteristics (funding, gender elements, project type, agenda) as controls. Model 4 includes the officer's position as a control variable. Model 5 – the full model, also includes country-level controls – namely the Gross Domestic Product of the country (GDP – logged) and population (logged). Model 6 presents results from entropy balancing estimations.

Table 3 (models 1 through 4) shows that women are positively, but not significantly associated with the relevance measure of project performance. The results from the full model and entropy matching (models 5 and 6) show a positive and significant association of women officers with project relevance. Specifically, women officers are associated with an increase in odds of an improvement in relevance rating by a factor of 1.32 in the full model and a factor of 1.32 when entropy matched. These results are significant at the 90% CI.

In Table 4, as I include the officer's position and country-level characteristics as controls, the coefficient on project effectiveness is rendered insignificant. However, the association of women officers with the project's effectiveness rating remains consistently positive.

Table 5 presents the most interesting finding of the paper. Women officers are significantly associated with an increase in projects' efficiency ratings. In the full model (model 5) after controlling for project, individual, and country-level characteristics and including year, sector, and country fixed effects, I find that women are associated with an increase in odds of an improvement in relevance rating by a factor of 1.43 in the full model and a factor of 1.47 when entropy matched. These results are significant at the 95% CI.

Table 6 presents the results for women officers' association with sustainability ratings. In all models, we see that women are not associated with a project's sustainability rating. Surprisingly, as I control for country-level characteristics this association turns from positive to negative. However, these results are not significant.

Section 5 – Robustness Checks

There are a few limitations of the study that can negatively impact the robustness of the analysis –model misspecification, omitted variable bias, selection bias, and bias in ratings. I employ several robustness checks to address the above issues where results are significant (women's positive and significant association with the relevance and efficiency ratings of project performance).

To address the model misspecification concern, I re-estimate equations (i) and (iii) using least square regressions. Results are presented in Table 7 and Table 8. Women officers' positive and significant association with efficiency ratings is consistent with findings in ordered logit models. However, in model 5- the full model, the positive association with the relevance ratings is rendered insignificant.

To test for omitted variable bias, I include other variables that were not included in the main analysis due to multicollinearity issues or more than 50% missing data. These variables are the human development index (Heinzel, 2022) and the gender of the director who approved the project.

The concern of self-selection arises if women or men self-select into certain types of projects. To account for self-selection – I have already included sector-fixed effects wherein women and men could be self-selecting into certain types of sectors. Additionally, I control for complexity of the project. Heinzel (2022) and Malik & Stone

(2018) code the number of objectives of a project to define the complexity of the World Bank project. I use the variable “report type” to account for complexity. Longer and more complex projects have “Project Completion Reports” – while smaller and less complex projects have “Technical assistance Completion reports.”.

For relevance, the results are presented in Table 9. Controlling for the complexity of the project in model 2 does not change the results. However, control for the human development index, and the gender of the director, the coefficient is rendered insignificant. It is important to note that due to missing data on the gender of the director, the number of observations was reduced by approximately 50%. Therefore, the results in model 5 are most likely imprecise and questionable – and might not be attributed to omitted variable bias. For efficiency, results are presented in Table 10. In all models (2-5), women officers are positively and significantly associated with the efficiency criteria of performance.

To address the bias in ratings, I take from the literature to show that staff at ADB have no incentive to game the ratings (Heinzel, 2022). Moreover, if women staff or men staff would bias the ratings, they would bias all elements of the performance rating, not just the efficiency and relevance ratings. As I show in Table 3, Table 4, Table 5, and Table 6 – women officers are positively and significantly only associated with efficiency and relevance ratings and not sustainability ratings.

Section 6 – Conclusion

This paper looks at the association of gender identity with project performance in aid organizations – specifically at a regional development bank – ADB. This paper draws from the interdisciplinary literature of development, financial economics, management, and the bureaucratic representation theory to pose a central question – Do men and

women leaders at ADB produce different outcomes on the relevance, effectiveness, efficiency, and sustainability criteria of project performance? Using multinomial logit regressions, and conducting various robustness checks, I find that women officers are associated with significantly positive outcomes on the efficiency criterion of project performance. They are also significantly associated with the relevance criterion of project performance, but this association is subject to model specification. Women officers are positively (but not significantly) associated with the effectiveness and sustainability criterion of project performance.

This paper contributes to the literature by looking at how identity attributes – specifically gender matters for a context other than corporate management, or domestic grassroots level firms and organizations. It looks at a less researched but the largest regional development bank which is different from other international banks like the World Bank in their staffing and operations. This paper also breaks down performance measures to understand how the gender of project officers matters for different outcomes relative to the commonly used aggregate measure of project performance.

The key takeaway from this paper is that different measures of project performance are differently impacted by the gender of managers. Women are significantly associated with the efficiency rating of project performance. ADB defines efficiency as the economic use of resources for the country and society where the project is carried out and at the operational level. There are several reasons why women might be positively associated with the efficiency measure. 1) Research by Duflo (2000) and Thomas (1990) shows that compared to income or assets in the hands of men, income, or assets in the hands of women are associated with larger improvement in societal benefits, specifically child's health and nutrition. Women have also been shown

to have longer time horizons when planning resource allocations (Rubalcava et al., 2009). Therefore, at aid organizations where the use of resources leads to large-scale development impact, women might be using resources more efficiently. 2) We could be seeing a selection effect where women who do become managers are more efficient than men who become managers because it was relatively more difficult for them to get there. This selection effect can also be seen in recent work by Kandpal et al. (2023) where they show that men at international financial institutions (IFIs) are disproportionately likely to make it out of professional ranks into the managerial ones. Table 1 supports this argument by showing that women are underrepresented in managerial roles (30% of managerial roles are held by women relative to 70% by men). While the data shows clear underrepresentation of women managers, it does not show that women officers are less qualified than men officers. At the same time, if it is a selection effect at work, we might have also seen it for other measures of project performance – namely relevance, effectiveness, and sustainability. Therefore, more research into the mechanisms is warranted to understand why women produce better outcomes on the efficiency outcomes relative to men. Further qualitative work is needed to understand how men and women rate themselves on these performance criteria.

Three policy implications can be drawn from this paper. First, women are underrepresented in managerial roles at ADB. Hiring and promoting more women is not only a contribution towards the SDG goals of gender equality and achieving parity – but also good for performance. Second, is to realize that if we were to look only at an aggregate measure of project performance as has been done previously, we might be underestimating the role of women managers in the organization. Therefore, how data on performance is measured and analysed is extremely important. Third is to understand that context and levels of analysis matter when evaluating project performance. External

validity issues can be severe when generalizing results from corporate organizations, public sector organizations, and even international multilateral banks. Therefore, cases for research on performance should be carefully selected and expanded for 1) a holistic understanding of various factors associated with aid-project performance and 2) to tweeze out mechanisms that correlate gender identity with measures of performance.

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Tables

Table 1 - Summary Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
Relevance	1472	2.33	0.57	0	3
Effectiveness	1334	1.8	0.62	0	3
Efficiency	1365	1.88	0.63	0	3
Sustainability	1093	2.04	0.49	0	3
PO Gender (woman=1)	3456	0.3	0.46	0	1
PO quality	2915	2.97	0.35	2	4
Director Gender (woman =1)	646	0.3	0.46	0	1
Ratio of women/total staff	1312	0.26	0.23	0	1
GDP (log)	2759	7.28	0.92	4.92	10.53
Population (log)	2760	17.92	2.61	9.17	21.06
Funding (in million)	2878	42.5	108	0.005	1250
Project delays (in weeks)	3189	83.08	72.43	-304	548
Approval year	3470	2008.9	4.33	1998	2019
Closing year	3220	2012.8	4.12	2000	2020

The table presents summary statistics for relevant project staff and country characteristics (treated as continuous) for all projects at the Asian Development Bank.

Table 2 - Summary statistics for selected categorical variables.

<i>Selected Variables</i>	Women (N=997)	Men (N=2428)	Overall
<i>Relevance</i>			
Irrelevant	1 (0.1%)	0 (0%)	1
Less than Relevant	19 (1.9%)	50 (2.1%)	69
Relevant	267 (26.8%)	574 (23.6%)	841
Highly Relevant	178 (17.9%)	378 (15.6%)	556
Missing	532 (53.4%)	1426 (58.7%)	1958
<i>Effectiveness</i>			
Ineffective	11 (1.1%)	31 (1.3%)	42
Less than Effective	67 (6.7%)	162 (6.7%)	229
Effective	307 (30.8%)	618 (25.5%)	925
Highly Effective	59 (5.9%)	105 (4.3%)	164
Missing	553 (55.5%)	1512 (62.3%)	2065
<i>Efficiency</i>			
Inefficient	11 (1.1%)	37 (1.5%)	48
Less than Efficient	85 (8.5%)	184 (7.6%)	269
Efficient	299 (30.0%)	610 (25.1%)	909
Highly Efficient	33 (3.3%)	70 (2.9%)	103

Missing	569 (57.1%)	1527 (62.9%)	2096
<i>Sustainability</i>			
Unlikely	8 (0.8%)	29 (1.2%)	37
Less than Likely	0 (0%)	0 (0%)	0
Likely	294 (29.5%)	641 (26.4%)	935
Most Likely	42 (4.2%)	76 (3.1%)	118
Missing	653 (65.5%)	1682 (69.3%)	2335
<i>Gender Elements</i>			
Gender equity	131 (13.1%)	179 (7.4%)	310
Effective gender mainstreaming	127 (12.7%)	323 (13.3%)	450
Some gender elements	144 (14.4%)	416 (17.1%)	560
No gender elements	338 (33.9%)	866 (35.7%)	1204
Missing	257 (25.8%)	644 (26.5%)	901
<i>Completion Report Types</i>			
Project Completion Report	240 (24.1%)	564 (23.2%)	804
Technical Assistance Completion Report	482 (48.3%)	1048 (43.2%)	1530
Missing	275 (27.6%)	816 (33.6%)	1091
<i>Countries with the most projects</i>			
China	153 (15.3%)	341 (14.0%)	494
India	58 (5.8%)	236 (9.7%)	294
Viet Nam	52 (5.2%)	145 (6.0%)	197
Bangladesh	34 (3.4%)	128 (5.3%)	162
Pakistan	34 (3.4%)	126 (5.2%)	160
<i>Countries with the least projects</i>			
Malaysia	2 (0.2%)	0 (0%)	2
Brunei Darussalam	1 (0.1%)	2 (0.1%)	3
Palau	0 (0%)	3 (0.1%)	3
Turkmenistan	3 (0.3%)	3 (0.1%)	6
Nauru	0 (0%)	6 (0.2%)	6
<i>Sector</i>			
Agriculture and Natural Resources	163 (16.3%)	455 (18.7%)	618
Education	110 (11.0%)	125 (5.1%)	235
Energy	113 (11.3%)	373 (15.4%)	486
Finance	85 (8.5%)	229 (9.4%)	314
Health	59 (5.9%)	76 (3.1%)	135
Industry and trade	45 (4.5%)	87 (3.6%)	132
Information and communication technology	8 (0.8%)	20 (0.8%)	28
Multisector	12 (1.2%)	12 (0.5%)	24
Public sector management	228 (22.9%)	428 (17.6%)	656
Transport	96 (9.6%)	440 (18.1%)	536
Water and other urban infrastructure and services	78 (7.8%)	183 (7.5%)	261

The table presents summary statistics for relevant categorical variables for all projects at the Asian Development Bank.

Table 3 - regressing gender of project officer on relevance

	(1)	(2)	(3)	(4)	(5)	(6)
	Relevance	Relevance	Relevance	Relevance	Relevance	Relevance
Estimation approach	ologit	ologit	ologit	ologit	ologit	ologit (Balancing)
Women officer	1.12 (0.12)	1.19 (0.14)	1.26 (0.19)	1.16 (0.18)	1.32* (0.22)	1.38* (0.23)
Funding			1* (0.00)	1* (0.00)	1* (0.00)	
Gender elements			0.91 (0.08)	0.86 (0.18)	0.83 (0.09)	
Project type			Yes	Yes	Yes	Yes
Project agenda			Yes	Yes	Yes	Yes
Officer position				Yes	Yes	Yes
GDP (log)					1.2 (0.50)	
Population (log)					1.1 (0.27)	
Country fixed effects		Yes	Yes	Yes	Yes	Yes
Approval year fixed effects		Yes	Yes	Yes	Yes	Yes
Closing year fixed effects		Yes	Yes	Yes	Yes	Yes
Sector fixed effects		Yes	Yes	Yes	Yes	Yes
/						
cut1	- 7.247*** (1.00)	- 7.490*** (1.60)	- 6.768*** (1.91)	- 6.767*** (2.02)	- 2.773 (3.74)	- 5.650*** (1.76)
cut2	- 2.934*** (0.13)	- 3.130** (1.26)	-2.558 (1.67)	-2.470 (1.81)	1.542 (3.61)	-1.961 (1.52)
cut3	0.545*** (0.07)	0.729 (1.25)	1.388 (1.66)	1.618 (1.80)	5.614 (3.62)	2.208 (1.51)

Observations	1457	1457	1084	1084	1012	1012
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Significance levels - "*" $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$ ". Robust standard errors in brackets. Results are presented as odds ratios. This table regresses the gender of the project officer on the relevance criteria of project performance. Model 5 – the full model controls for project, individual, and project-level characteristics. Model 6 matches all control variables.

Table 4 - regressing gender of project officer on effectiveness

	(1)	(2)	(3)	(4)	(5)	(6)
	Effectiveness	Effectiveness	Effectiveness	Effectiveness	Effectiveness	Effectiveness
Estimation approach	ologit	ologit	ologit	ologit	ologit	ologit (Balancing)
Women officer	1.23* (0.15)	1.29* (0.18)	1.47** (0.24)	1.23 (0.22)	1.25 (0.24)	1.3 (0.24)
Funding			1 (0.00)	1 (0.00)	1 (0.00)	
Gender elements			0.92 (0.09)	0.87 (0.09)	0.79** (0.11)	
Project type			Yes	Yes	Yes	Yes
Project agenda			Yes	Yes	Yes	Yes
Officer position				Yes	Yes	Yes
GDP (log)					1.03 (0.34)	
Population (log)					1.05 (0.29)	
Country fixed effects		Yes	Yes	Yes	Yes	Yes
Approval year fixed effects		Yes	Yes	Yes	Yes	Yes
Closing year fixed effects		Yes	Yes	Yes	Yes	Yes
Sector fixed effects		Yes	Yes	Yes	Yes	Yes
/						
cut1	- 3.189*** (0.15)	- 6.587*** (1.13)	- 7.547*** (1.44)	- 8.560*** (1.69)	- -7.793* (4.20)	- 9.129*** (1.82)

cut2	1.078*** (0.08)	4.262*** (1.12)	5.229*** (1.42)	6.168*** (1.67)	-5.443 (4.21)	6.618*** (1.80)
cut3	2.586*** (0.12)	-0.190 (1.11)	-0.866 (1.41)	-1.622 (1.67)	-0.880 (4.21)	-1.882 (1.78)
Observations	1319	1319	1007	1007	946	946

Significance levels - "*" p<0.10 ** p<0.05 *** p<0.01". Robust standard errors in brackets. Results are presented as odds ratios. This table regresses the gender of the project officer on the effectiveness criteria of project performance. Model 5 – the full model controls for project, individual, and project-level characteristics. Model 6 matches all control variables.

Table 5 - regressing gender of project officer on efficiency

	(1) Efficiency	(2) Efficiency	(3) Efficiency	(4) Efficiency	(5) Efficiency	(6) Efficiency
Estimation approach	ologit	ologit	ologit	ologit	ologit	ologit (Balancing)
Women officer	1.42*** (0.17)	1.47*** (0.2)	1.64*** (0.27)	1.41*** (0.24)	1.43** (0.25)	1.47** (0.27)
Funding			1** (0.00)	1** (0.00)	1** (0.00)	
Gender elements			0.94 (0.09)	0.9 (0.09)	0.87 (0.09)	
Project type			Yes	Yes	Yes	Yes
Project agenda			Yes	Yes	Yes	Yes
Officer position				Yes	Yes	Yes
GDP (log)					0.72 (0.18)	
Population (log)					1.38* (0.25)	
Country fixed effects		Yes	Yes	Yes	Yes	Yes
Approval year fixed effects		Yes	Yes	Yes	Yes	Yes
Closing year fixed effects		Yes	Yes	Yes	Yes	Yes

Sector fixed effects		Yes	Yes	Yes	Yes	Yes
/						
cut1	- 3.308*** (0.16)	- 4.646*** (1.22)	- 4.131*** (1.60)	- 5.095*** (1.71)	- 1.832 (3.06)	- 6.273*** (1.85)
cut2	- 1.256*** (0.08)	- 2.427** (1.22)	- -1.783 (1.59)	- -2.672 (1.70)	- 0.586 (3.06)	- 3.771** (1.83)
cut3	- 2.158*** (0.10)	- 1.438 (1.21)	- 2.247 (1.58)	- 1.482 (1.69)	- 4.700 (3.07)	- 0.416 (1.81)
Observations	1350	1350	1035	1035	968	968

Significance levels - "*" p<0.10 ** p<0.05 *** p<0.01". Robust standard errors in brackets. Results are presented as odds ratios. This table regresses the gender of the project officer on the efficiency criteria of project performance. Model 5 – the full model controls for project, individual, and project-level characteristics. Model 6 matches all control variables.

Table 6 - regressing gender of project officer on sustainability

	(1)	(2)	(3)	(4)	(5)	(6)
	Sustainability	Sustainability	Sustainability	Sustainability	Sustainability	Sustainability
Estimation approach	ologit	ologit	ologit	ologit	ologit	ologit (Balancing)
Women officer	1.31 (0.23)	1.3 (0.25)	1.29 (0.30)	0.85 (0.23)	0.9 (0.25)	0.99 (0.28)
Funding			1 (0.00)	1 (0.00)	1 (0.00)	
Gender elements			0.81*** (0.12)	0.72*** (0.11)	0.63*** (0.10)	
Project type			Yes	Yes	Yes	Yes
Project agenda			Yes	Yes	Yes	Yes
Officer position				Yes	Yes	Yes
GDP (log)					0.96 (0.47)	
Population (log)					1.16 (0.39)	

Country fixed effects		Yes	Yes	Yes	Yes	Yes
Approval year fixed effects		Yes	Yes	Yes	Yes	Yes
Closing year fixed effects		Yes	Yes	Yes	Yes	Yes
Sector fixed effects		Yes	Yes	Yes	Yes	Yes
/						
cut1	3.258*** (0.18)	5.382*** (1.64)	5.626*** (2.17)	5.432** (2.46)	-2.667 (4.80)	-4.478* (2.66)
cut2	2.200*** (0.12)	1.491 (1.62)	1.700 (2.18)	2.322 (2.49)	5.151 (4.84)	3.719 (2.68)
Observations	1083	1083	847	847	794	794

Significance levels - "*" p<0.10 ** p<0.05 *** p<0.01". Robust standard errors in brackets. Results are presented as odds ratios. Results are presented as odds ratios. This table regresses the gender of the project officer on the sustainability criteria of project performance. Model 5 – the full model controls for project, individual, and project-level characteristics. Model 6 matches all control variables.

Table 7 - robustness check for model misspecification - regressing gender of project officer on relevance

	(1) Relevance	(2) Relevance	(3) Relevance	(4) Relevance	(5) Relevance	(6) Relevance
Estimation approach	ols	ols	ols	ols	ols	ols (Balancing)
Women officer	0.036 (0.03)	0.044 (0.03)	0.053 (0.04)	0.031 (0.04)	0.057 (0.04)	0.068* (0.04)
Funding			0.000** (0.00)	0.000** (0.00)	0.000** (0.00)	
Gender elements			-0.025 (0.02)	-0.039* (0.02)	0.047* (0.03)	
Project type			Yes	Yes	Yes	Yes
Project agenda			Yes	Yes	Yes	Yes
Officer position				Yes	Yes	Yes

GDP (log)					0.021 (0.09)	
Population (log)					0.031 (0.06)	
Country fixed effects		Yes	Yes	Yes	Yes	Yes
Approval year fixed effects		Yes	Yes	Yes	Yes	Yes
Closing year fixed effects		Yes	Yes	Yes	Yes	Yes
Sector fixed effects		Yes	Yes	Yes	Yes	Yes
Constant	2.315*** (0.02)	3.262*** (0.31)	2.246*** (0.48)	2.118*** (0.48)	2.226** (0.87)	2.043*** (0.44)
Observations	1457	1457	1084	1084	1012	1012

Significance levels - "*" $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$ ". Robust standard errors in brackets. Results are presented as beta coefficients. Model 5 presents the full model and Model 6 matches on control variables.

Table 8 - robustness check for model misspecification - regressing gender of project officer on efficiency

	(1) Efficiency	(2) Efficiency	(3) Efficiency	(4) Efficiency	(5) Efficiency	(6) Efficiency
Estimation approach	ols	ols	ols	ols	ols	ols (Balancing)
Women officer	0.110*** (0.03)	0.106*** (0.04)	0.136*** (0.04)	0.088** (0.04)	0.092** (0.05)	0.093** (0.05)
Funding			0.000** (0.00)	0.000** (0.00)	0.000** (0.00)	
Gender elements			-0.017 (0.03)	-0.028 (0.03)	-0.038 (0.03)	
Project type			Yes	Yes	Yes	Yes
Project agenda			Yes	Yes	Yes	Yes
Officer position				Yes	Yes	Yes
GDP (log)					-0.095	

					(0.07)	
Population (log)					0.081*	
					(0.05)	
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Approval year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Closing year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Sector fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Constant	1.844***	2.132***	1.441***	1.679***	0.943	1.658***
	(0.02)	(0.34)	(0.41)	(0.42)	(0.84)	(0.52)
Observations	1350	1350	1035	1035	968	968

Significance levels - "*" $p < 0.10$ ** $p < 0.05$ *** $p < 0.01$ ". Robust standard errors in brackets. Results are presented as beta coefficients. Model 5 presents the full model and Model 6 matches on control variables.

Table 9 - robustness check for omitted variable bias and self-selection bias - regressing gender of project officer on relevance

	(1)	(2)	(3)	(4)	(5)
	Relevance	Relevance	Relevance	Relevance	Relevance
Estimation approach	ologit	ologit	ologit	ologit	ologit
Woman officer	1.32*	1.32*	1.3	1.3	0.93
	(0.22)	(0.22)	(0.22)	(0.22)	(0.27)
<i>Report Type</i>					
Project Completion Report		1.105		1.03	1.47
		(.40)		(0.39)	(1.00)
TA Completion Report		0.000		0.000	0.000
		(.)		(.)	(.)
HDI Index			0.14	0.14	0.4
			(1.38)	(1.36)	(6.97)
Woman Director					1.45
					(0.40)
Funding	1*	1**	1*	1*	1
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Gender Elements	0.83	0.83	0.84*	0.84*	0.94

Project type	(0.09) Yes	(0.09) Yes	(0.09) Yes	(0.09) Yes	(0.17) Yes
Project agenda	Yes	Yes	Yes	Yes	Yes
Officer position	Yes	Yes	Yes	Yes	Yes
GDP (log)	1.2 (0.50)	1.2 (0.50)	1.14 (0.52)	1.14 (0.52)	2.08 (1.51)
Population (log)	1.1 (0.27)	1.1 (0.27)	1.16 (0.32)	1.16 (0.32)	0.62 (0.45)
Country fixed effects	Yes	Yes	Yes	Yes	Yes
Approval year fixed effects	Yes	Yes	Yes	Yes	Yes
Closing year fixed effects	Yes	Yes	Yes	Yes	Yes
Sector fixed effects /	Yes	Yes	Yes	Yes	Yes
cut1	- 2.773 (3.74)	- 2.725 (3.75)	- 3.055 (6.24)	- 3.046 (6.23)	- 3.195 (16.73)
cut2	1.542 (3.61)	1.590 (3.62)	1.267 (6.20)	1.276 (6.19)	1.149 (16.74)
cut3	5.614 (3.62)	5.662 (3.63)	5.348 (6.20)	5.357 (6.19)	
Observations	1012	1012	999	999	536

Significance levels - "*" p<0.10 ** p<0.05 *** p<0.01". Robust standard errors in brackets. Results are presented as odds ratios. Model 1 is the full model – equivalent to model 5 in Table 3. Model 2 includes Report Type as controls. Model 3 includes the HDI index. Model 4 includes both the Report type and HDI index. Model 5 includes the gender of ADB director during the time of project approval.

Table 10 - robustness check for omitted variable bias and self-selection bias - regressing gender of project officer on efficiency.

	(1)	(2)	(3)	(4)	(5)
	Efficiency	Efficiency	Efficiency	Efficiency	Efficiency
Estimation approach	ologit	ologit	ologit	ologit	ologit
Woman officer	1.43** (0.25)	1.43** (0.25)	1.42* (0.26)	1.42* (0.26)	1.73* (0.52)
<i>Report Type</i>	No		No		
Project Completion Report		1.05 (0.37)		1.1 (0.40)	1.21 (0.75)
TA Completion Report		0.000 (.)		0.000 (.)	0.000 (.)
HDI Index	No	No	0.09 (0.79)	0.1 (0.87)	0.002 (0.03)
Woman Director	No	No	No	No	2.19*** (0.62)
Funding	1** (0.00)	1** (0.00)	1** (0.00)	1** (0.00)	1 (0.00)
Gender Elements	0.87 (0.09)	0.87 (0.09)	0.86 (0.09)	0.87 (0.09)	1.17 (0.19)
Project type	Yes	Yes	Yes	Yes	Yes
Project agenda	Yes	Yes	Yes	Yes	Yes
Officer position	Yes	Yes	Yes	Yes	Yes
GDP (log)	0.72 (0.18)	0.72 (0.19)	0.74 (0.20)	0.74 (0.20)	0.48 (0.27)
Population (log)	1.38* (0.25)	1.39* (0.25)	1.40* (0.27)	1.39* (0.27)	2.04 (1.39)
Country fixed effects	Yes	Yes	Yes	Yes	Yes
Approval year fixed effects	Yes	Yes	Yes	Yes	Yes
Closing year fixed effects	Yes	Yes	Yes	Yes	Yes
Sector fixed effects	Yes	Yes	Yes	Yes	Yes

/					
cut1	-1.832	-1.811	2.533	2.448	-2.823
	(3.06)	(3.05)	(5.41)	(5.40)	(16.57)
cut2	0.586	0.607	0.123	0.038	-0.340
	(3.06)	(3.05)	(5.42)	(5.41)	(16.58)
cut3	4.700	4.721	3.979	4.064	4.272
	(3.07)	(3.07)	(5.43)	(5.42)	(16.58)
Observations	968	968	955	955	534

Significance levels - "*" p<0.10 ** p<0.05 *** p<0.01". Robust standard errors in brackets. Results are presented as odds ratios. Model 1 is the full model – equivalent to model 5 in table 5. Model 2 includes Report Type as controls. Model 3 includes the HDI index. Model 4 includes both the Report type and HDI index. Model 5 includes the gender of ADB director during the time of project approval.

Figures

Figure 1 - Distribution of performance ratings. The ratings do not follow a normal distribution.



Figure 2 – approximately 30% of managers at ADB are women and 70% are men. Women are underrepresented as project officers in ADB.



Figure 3 - Ratings by Gender. Women are positively associated with all measures of project performance.

